

FOR REFERENCE

2009/867/2

**GUJARAT UNIVERSITY
BE SEM. III CE EXAMINATION
DATA STRUCTURE & ALGORITHM**

Date:

Time:

Max. Marks :100

Instruction : 1) Right side figure indicates marks.

2) Each section must be written in separate answer sheet.



SECTION -I

Q.1 (a) Write an algorithm to insert a node into linked list at last position. [6]

(b) Explain various operation of circular queue. [6]

(c) Create binary search tree for following value. [6]
15,25,30,5,10,8,32,34,7,6,28.

Q.2 (a) Trace bubble sort for following values. [7]
55,30,35,25,30,2011,15.

(b) What is searching ? Write a program for binary search. [7]

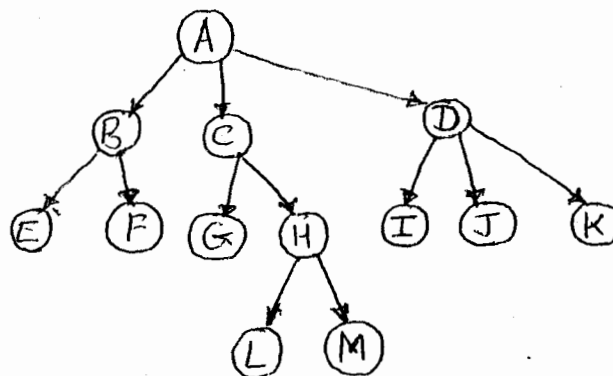
OR

Q.2 (a) Define heap tree. Explain creation of heap tree with example. [7]

(b) Explain radix sort with example. [7]

Q.3 Attempt any three. [18]

(a) Convert following General tree into binary tree.



(b) Write a short note on threaded binary tree.

(c) Convert following expression into prefix notation.

(i) $p + (q + r) * s$ (ii) $p \wedge q * r \setminus s$ (iii) $(p + q) \setminus r * s$.

(d) Write an algorithm for checking balanced parenthesis of expression.

(e) Write a short note on priority queue.

SECTION -II

Q.4 (a) Define : i) Graph ii) Weighted Graph iii) Sparse matrix
iv) Reachability matrix

[8]

(b) What is traversal ? Which are various traversal method of tree. Explain any one with example..

[5]

(c) Write a recursive and non-recursive program for finding factorial of given number . Which is better one ? How?

[5]

Q.5 (a) Write and explain shortest path algorithm.

[7]

(b) Write a recursive program to create binary tree.

[7]

OR

Q.5 (a) Explain weight balance tree with example.

[7]

(b) What is hash collision? Explain various hash collision resolution techniques.

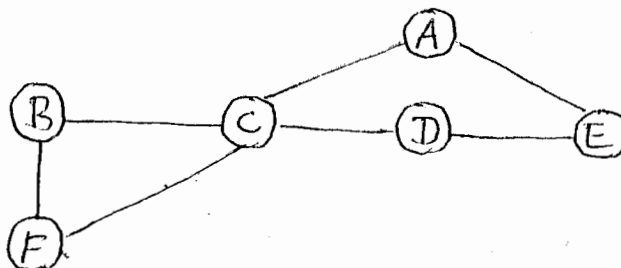
[7]

Q.6 Attempt any three of following question.

[18]

(a) Write a DFS traversal algorithm.

(b) Draw DFS traversal for following graph.



(c) Write a short note on Multi list file organization.

(d) Explain procedure to find path matrix from adjacency matrix with example.

(e) Write an algorithm for addition of two polynomials.

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